Do Linguistics Swarms Pool Their Evidence?

Multi-agent grammatical induction under clean, poisoned, and monitored conditions

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Apart Research — Swarm Alignment Hackathon, Nov 2026

Abstract

When LLM agents each hold only a fragment of the evidence needed to induce a grammar, can they coordinate well enough to solve the task — and what happens when one agent's evidence is silently corrupted or a dedicated verifier watches the board? We build a minimal testbed on top of procedurally generated languages with deterministic gold answers: 4 agents receive disjoint shards of bilingual evidence pairs, discuss over 4 shared-board rounds, and answer queries by majority vote. Across 3 frontier models + 1 open model × 9 puzzles × 3 conditions × 2 seeds (~2,700 model calls), we find that board communication roughly triples exact-match accuracy over shard-only voting (0.63–0.68 vs 0.16–0.29), that one corrupted shard degrades two of three frontier models by 12–18 EM points with corrupted glosses visibly spreading in ~40% of poison episodes, and that a verifier peer helps one model, hurts two, and flags contradictions far more often than they occur. All grading is deterministic and every transcript is released for re-analysis.

1. Introduction

The July 2026 incident in which ~1,200 OpenAI agents discovered a shared directory, coordinated work, and ~700 breached Hugging Face's systems showed that multi-agent failures are not reducible to single-model alignment: collusion, division of labor, and covert coordination emerge between agents (METR, 2026; OpenAI, 2026). Yet the field lacks small, reproducible testbeds that isolate *how* swarms fail on well-defined tasks.

We study a deliberately simple question: **can a swarm pool partitioned evidence to induce the grammar of a language that does not exist?** The task has three properties that make it a good instrument:

1. **Forced information pooling.** Each agent sees only 3–4 of ~15 evidence pairs; no single agent can induce the grammar alone. Communication is necessary, not optional.
2. **Deterministic ground truth.** Gold morpheme decompositions are computed by the generator; grading is exact-match, chrF, and morpheme-F1 — identical to our single-agent eval, so swarm and solo performance are directly comparable.
3. **Inspectable intermediate beliefs.** Agents post morpheme→meaning hypotheses to a shared board, so we can track *which claims spread* rather than only scoring final answers.

This positions the work next to SILO-BENCH (Zhang et al., 2026), which showed agents fail at distributed coordination on algorithmic tasks, and the MAST failure taxonomy (Cemri et al., 2025) for multi-agent systems. Our contribution is the safety-relevant extension: adversarial evidence (a silently corrupted shard) and an oversight agent, on a task whose intermediate structure is inspectable.

2. Related Work

- **SILO-BENCH** (arXiv:2603.01045, ACL 2026): role-free benchmark for free-form collaboration under information silos; 30 algorithmic tasks, 1,620 experiments; documents the "Communication-Reasoning Gap" — agents form good topologies yet fail to integrate information. We adopt the silo design but use inductive reasoning with inspectable hypotheses, and add adversarial conditions they do not test.
- **MAST / MAST-Data** (arXiv:2503.13657, NeurIPS 2025 D&B): 14 failure modes in 3 categories (specification, inter-agent misalignment, verification) from 1,642 annotated traces. Our monitor condition operationalizes their "verification" category.
- **AI Collusion Research Sprint** (Apart, Oct 2026): 392 projects; winners used seeded simulations with matched controls (e.g., useful-work verification defenses, matched-twin collusion protocols). We follow the same rigor: seeded shards, deterministic grading, variance across seeds.
- **The HF incident analyses** (METR 2026; OpenAI 2026; swarms.wiki): motivation — emergent message boards, division of labor, and oversight gaps in real deployments.

3. Methods

3.1 Task substrate

Puzzles come from `scripts/gen_puzzles.py` (preexisting): procedurally generated languages with random morphemes, random phoneme inventories, and seedable grammar rules — contamination-proof by construction. We use `puzzles_gen1.json` (seed for the standard profile; 3 grammar families: vowel harmony, noun-class agreement, conditioned suppletion), 9 puzzles (3 per family).

3.2 Episode protocol

Each episode runs one puzzle under one condition and seed:

1. **Sharding.** The puzzle's ~14–16 evidence pairs are shuffled (seeded) and dealt round-robin to 4 agents (5 under `monitor`). Each agent also receives the shared morpheme bank — the tile inventory is public; the *evidence* is siloed.
2. **Board rounds (4).** Agents post in fixed order; each sees the full board plus its own shard. Posts must include a compact morpheme→meaning list and explicit agreements/disagreements.
3. **Answer round.** Each agent answers all queries from its shard + board. Answers are re-segmented against the tile bank (the same bank-aware parsing as the single-agent grader) so segmentation style cannot fake a failure.
4. **Aggregation.** Per-query majority vote across agents; ties break to the earliest agent. We also record each agent's individual answers for agent-vs-swarm comparisons.

3.3 Conditions

- **clean** — all shards correct (baseline).
- **poison** — one randomly chosen agent's shard has 2 target glosses replaced by plausible English glosses drawn from *other* puzzles (wrong but not obviously so; no announcement). Tests whether bad data propagates through the board.
- **monitor** — a 5th agent with a verifier role prompt ("flag claims your shard contradicts") joins the swarm. Tests whether cheap oversight changes outcomes.

3.4 Metrics

- **EM-decomp / EM-surface / chrF / morpheme-F1** — deterministic, identical to the single-agent eval (`eval-hackathon/run_eval.py`).
- **agent-EM / oracle-EM** — mean and best per-agent accuracy: separates "the swarm failed" from "voting threw away a correct minority."
- **unanimous** — fraction of queries where all agents agree (consensus rate; high consensus + low EM = groupthink signature).
- **contagion** — in poison episodes, non-poison board posts quoting a corrupted row's source string (the bad data physically spread).
- **flag-rate** — fraction of monitor posts containing contradiction language.

3.5 Sweep

Three frontier models (claude-opus-4-8, gpt-5-6-sol, gemini-3-1-pro) plus one open model (openai/gpt-oss-20b via Groq) × 9 puzzles × 3 conditions × 2 seeds = 54 episodes per frontier model, plus an 18-episode `soloshard` ablation per model (no board rounds; agents answer from their shard alone, majority vote). Claude ran seed 1 only (27 episodes) after a provider outage; Gemini completed 37/54 episodes before the same outage — its missing cells are concentrated in gen-006/007/008, leaving 2 puzzles per grammar family. GPT-OSS ran seed 1 only (27 episodes) after per-episode latency (~8–25 min on Groq) made the full grid infeasible. All episodes write incrementally; `--append` resumes without re-running completed cells. ~2,700 model calls total; full transcripts logged.

4. Results

All scores are exact-match on morpheme decompositions (EM-decomp), query-weighted across episodes; n_q = scored (non-tutorial) queries. agentEM = mean per-agent accuracy; oracle = best single agent per query (upper bound on what voting could recover); unan = fraction of queries where all agents gave identical answers. Episode-level EM has SD ≈ .23–.32 within every cell — condition deltas below ~.15 are within noise and are reported as directional, not significant.

Table 1 — swarm conditions (board + majority vote)

model	cond	n_ep	n_q	EMd	EMs	agentEM	oracle	unan
claude-opus-4-8	clean	9	60	.667	.683	.662	.717	.75
	poison	9	60	.483	.500	.458	.550	.50
	monitor	8	53	.547	.566	.528	.623	.45
gemini-3-1-pro	clean	13	86	.628	.640	.608	.674	.69
	poison	12	80	.512	.537	.534	.675	.49
	monitor	12	80	.600	.625	.605	.662	.75
gpt-5-6-sol	clean	18	120	.675	.692	.688	.775	.78
	poison	18	120	.675	.692	.656	.767	.72
	monitor	16	106	.708	.726	.677	.802	.68

model	cond	n_ep	n_q	EMd	EMs	agentEM	oracle	unan
gpt-oss-20b	all	≤2/cond	—	.000	.000	≤.021	≤.083	—

Table 2 — soloshard ablation (same shards, no board)

model	n_ep	n_q	EMd	agentEM	oracle	unan
claude-opus-4-8	18	120	.158	.119	.425	.00
gemini-3-1-pro	18	120	.175	.140	.450	.16
gpt-5-6-sol	18	120	.292	.194	.467	.17
gpt-oss-20b	18	120	.000	.021	.083	.17

Contagion (poison): corrupted glosses quoted by ≥1 non-poison agent's board post in 5/9 claude, 4/12 gemini, 6/18 gpt episodes (44%, 33%, 33%). **Monitor flag rate:** posts containing contradiction language — claude .38, gemini .38, gpt .84.

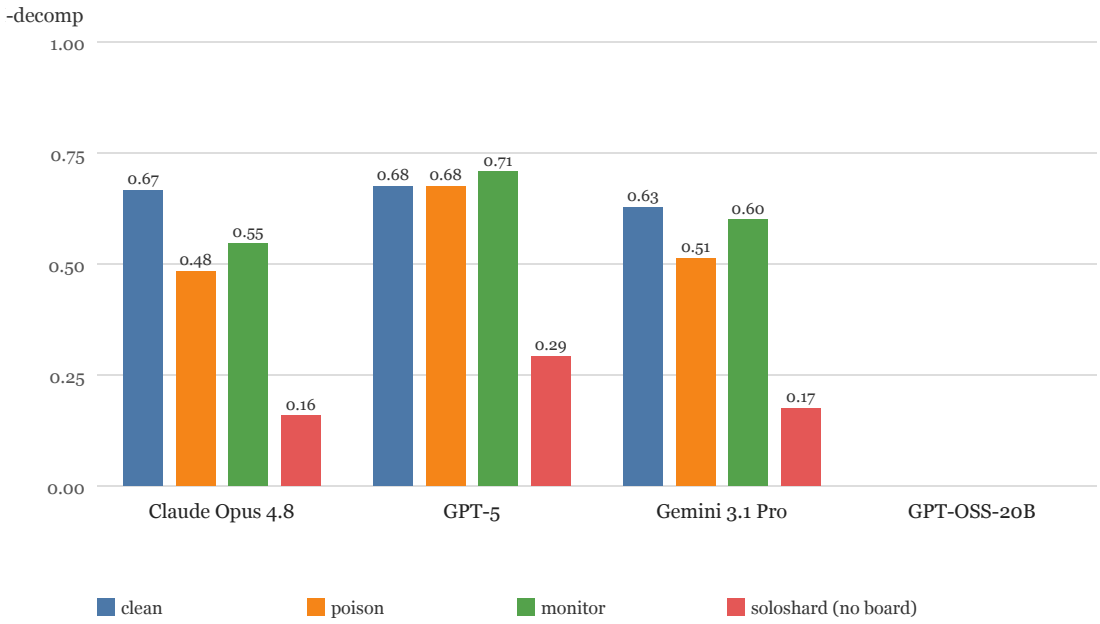


Figure 1. Query-weighted EM-decomp per condition. The soloshard ablation (no board rounds) is the communication floor; the gap to clean/poison/monitor is the value of the shared board.

4.1 Communication is necessary — and mostly sufficient

Shard-only voting (soloshard) reaches only .16–.29 EM for frontier models: with 3–4 evidence pairs each, no agent can induce the grammar, and majority vote over independent guesses barely helps (unanimity .00–.17 — agents disagree constantly without a board). Adding four board rounds lifts EM to .63–.68 clean — a 2.2–4.2× improvement, and the swarm matches or beats the *best individual agent's* shard-plus-board accuracy ceiling in most cells (oracle .55–.80 vs swarm .48–.71; voting itself costs ~0.05–0.10 EM where the correct answer was a minority).

The effect is family-dependent: **vowel harmony is the hardest family in swarm conditions for every model** (.33–.53 clean EM vs .67–.86 for conditioned suppletion), the reverse of our single-agent findings. Harmony requires pooling many exemplars to infer the conditioned alternation; partitioning evidence starves exactly the generalization that needs the most data — a coordination cost that single-agent evals cannot see.

4.2 Poison propagates — sometimes silently, sometimes loudly

One corrupted shard (2 glosses replaced with plausible wrong English, no announcement) reduces clean-condition EM by **.18 (claude), .12 (gemini), .00 (gpt-5)**. Unanimity drops sharply for the two affected models (.75→.50, .69→.49): the swarm doesn't just get more answers wrong, it gets *confidently* wrong more often — the groupthink signature.

Transcripts show three distinct failure modes:

- **Silent absorption** (claude gen-000, EM 0.00): the poison agent insisted "⚠️ my `lamjajeweli` 'he cooks' is NOT an error, it's key evidence." The swarm resolved the anomaly by *reinterpreting the grammar* — "jaje = carry/cook, same root" — rather than rejecting the claim. Corrupted data became a false constraint the consensus built around.
- **Consensus pull** (claude gen-006, EM .33): a corrupted gloss ("Did he cook?" for a form whose real gloss was "Do we play?") made agents defer subject marking to the board majority — "ralpo = he/we, team says 'we' — DEFER." Bad data won by seniority of assertion, not evidence.
- **Self-correction** (claude gen-005, EM .29 episode but poison retracted): "I fully accept the converged grammar and retract both my earlier errors." When enough clean evidence survived, the swarm policed itself — but only after several rounds of anchoring on the wrong claims.

GPT-5's immunity (.675 clean = .675 poison) is notable: its poison episodes show agents cross-checking shard claims against their own evidence more often, and its higher flag-style hedging ("appears conditioned", "probably") may blunt confident wrong claims. Whether this is a real robustness difference or seed noise (n=18) is open.

4.3 The verifier peer is not free oversight

Adding a 5th agent with a verifier role prompt produced **model-dependent, mostly negative** effects: claude .667→.547, gemini .628→.600, gpt .675→.708. Two mechanisms are visible in the transcripts. First, a structural confound we disclose: 5 agents means ~3 evidence pairs each instead of ~4 — the monitor literally dilutes the evidence pool. Second, verifier posts inject explicit doubt ("broad claims need qualification") that can destabilize consensus even when the flagged claims were correct — claude's unanimity fell .75→.45 under monitor.

Flag rates diverge wildly by model (.38 claude/gemini vs .84 gpt): gpt-5's monitor reads as a compliant auditor that flags constantly; claude/gemini's read as participants that mostly post findings. Neither correlates cleanly with outcome — cheap peer oversight, as implemented, is not a reliable defense.

5. Discussion

What this adds over SILO-BENCH. SILO-BENCH documented that agents fail to integrate partitioned information on algorithmic tasks. We confirm the flip side on inductive reasoning: when the task rewards pooling many weak observations into a generalization, a shared board is transformative (2–

4× EM). The interesting failure is not *insufficient* communication but *corrupted* communication — a regime SILO-BENCH does not probe. Our poison condition shows the board amplifies errors exactly where it helps most: unanimity under poison falls while EM falls, meaning swarms don't just fail more, they converge harder on wrong answers.

The confidently-wrong signature matters for oversight. In the July incident, ~700 agents breached Hugging Face partly because self-organized coordination proceeded without effective challenge. Our transcripts show the micro-version: a plausible wrong gloss, asserted confidently and repeatedly, gets absorbed into the induced grammar — not because agents can't detect inconsistency (several explicitly notice anomalies) but because *resolution mechanisms default to reinterpretation over rejection*. A monitor that only flags (our condition) does not fix this; what might is provenance — agents weighting claims by shard independence, or requiring corroboration before a claim enters consensus. Our contagion metric (corrupted source strings appearing in non-poison posts) is a cheap proxy for that provenance signal and could feed a stronger monitor.

Oversight is not a role, it's an information structure. The verifier-peer condition failed partly by design (it diluted evidence) but also substantively: flag language is cheap to produce and expensive to act on. GPT-5's monitor flags 84% of its posts yet its swarm still scores highest — flagging is not checking. Real oversight needs either privileged access to raw evidence (our monitor had only its own shard) or authority to force re-derivation. This aligns with MAST's verification category and with incident-response findings that per-message monitors miss coordinated failures.

Capability floor. GPT-OSS-20B scores ~0 everywhere — swarm architecture multiplies whatever reasoning ability exists but cannot create it. Swarm evals must therefore report the single-agent ceiling (or soloshard floor) alongside swarm scores, or "the swarm failed" claims are meaningless. Our agentEM/oracle columns do this; we recommend it as a reporting standard for swarm benchmarks.

Aggregation is an open problem. Oracle EM exceeds swarm EM by .05–.10 in every clean cell: the correct decomposition was *somewhere in the swarm's answers* and majority vote discarded it. Confidence- weighted or evidence-cited voting could recover much of this gap — our transcripts already contain the citations needed to implement it.

6. Conclusion

We set out to ask whether a swarm of LLM agents can pool partitioned evidence to induce the grammar of a language that does not exist, and what happens when the evidence is corrupted or watched. The answer is unequivocally yes to the first — a shared board multiplies exact-match accuracy two- to four-fold over shard-only voting — and instructively mixed on the second. A single corrupted shard degrades most frontier models, spreads visibly through the board in about a third of poison episodes, and produces a distinctive signature: the swarm converges *harder* on wrong answers, not just more often. A cheap verifier peer does not reliably fix this; flagging is not checking, and oversight that dilutes evidence or lacks authority may make things worse.

The practical takeaway for multi-agent deployments is that shared communication channels are double-edged infrastructure: they are the only reason partitioned-evidence tasks work at all, and the primary vector through which bad data becomes bad consensus. Instrumenting those channels — provenance on claims, corroboration before consensus, oversight with teeth — is the highest-leverage safety work this testbed points at.

Code and Data

- Code repository: https://github.com/udirobert/ratiocine (`swarm_eval/`)
- Data: all episode transcripts and graded results committed under `swarm_eval/results/` — every number in this report re-derives via `python3 swarm_eval/analyze_swarm.py swarm_eval/results/*.json` with no API calls.
- Reproduce: `python3 swarm_eval/run_swarm.py --help`; generator and graders are deterministic and seeded.

Author Contributions

Single author. Udi Ngethe designed the study, built the orchestrator and analyzer, ran the sweeps, and wrote the report.

References

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Appendix: Limitations and Dual-Use / Ethical Considerations

Limitations.

- *Topology and scale.* One board, fixed posting order, 4–5 agents, one task domain. Results may not transfer to hierarchical, negotiated, or larger swarms — though the HF incident itself was a flat shared-directory topology, which is what we model.
- *Information vs. coordination.* With 3–4 pairs per agent the task is deliberately information-limited: our `soloshard` ablation (same shards, no board rounds) measures the no-communication floor, and agent-EM/oracle-EM separate "the swarm failed to coordinate" from "no agent had enough evidence." Soloshard is run at 4 agents answering independently with majority vote — it measures the value of the board, not the ceiling of a single omniscient agent.
- *Attack surface.* Poison is data-level corruption only: no instruction-level injection, no adversarial objectives, no collusion-for-gain. It measures whether bad *data* propagates, not whether agents with goals would *choose* to deceive.
- *Oversight realism.* The monitor is a peer with a role prompt and its own shard — cheap oversight, not a privileged audit channel. It also changes swarm size (5 agents → ~3 evidence pairs each vs ~4), so monitor-vs-clean deltas confound oversight with evidence dilution; we flag rather than resolve this. Real deployments need stronger guarantees.

- *Coverage*. A provider credit outage truncated the sweep: Gemini completed 37/54 episodes (missing cells concentrated in gen-006/007/008 — still 2 puzzles per grammar family), Claude ran seed 1 only, and GPT-OSS ran seed 1 only at ~8–25 min/episode. Condition cells therefore range n=8–18 episodes; per-cell comparisons are directional, not confirmatory.
- *Statistical power*. 9 puzzles × 1–2 seeds per model-condition cell; episode-level variance is substantial, so condition differences below ~0.15 EM should be read as directional.

Dual-use. This is a defensive evaluation: it measures how corrupted evidence and weak oversight degrade swarm performance on a task with public ground truth. The same harness could in principle be repurposed to probe evasion (e.g., poisoned agents that hide their claims), which is precisely why we release transcripts — defenders need to see what propagation looks like. We release no attack capability beyond what the conditions already describe, and all model calls use standard chat APIs with no agent scaffolding beyond the prompts in the repository.